Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

ETHEREUM FRAUD DETECTION VIA SUPERVISED LEARNING

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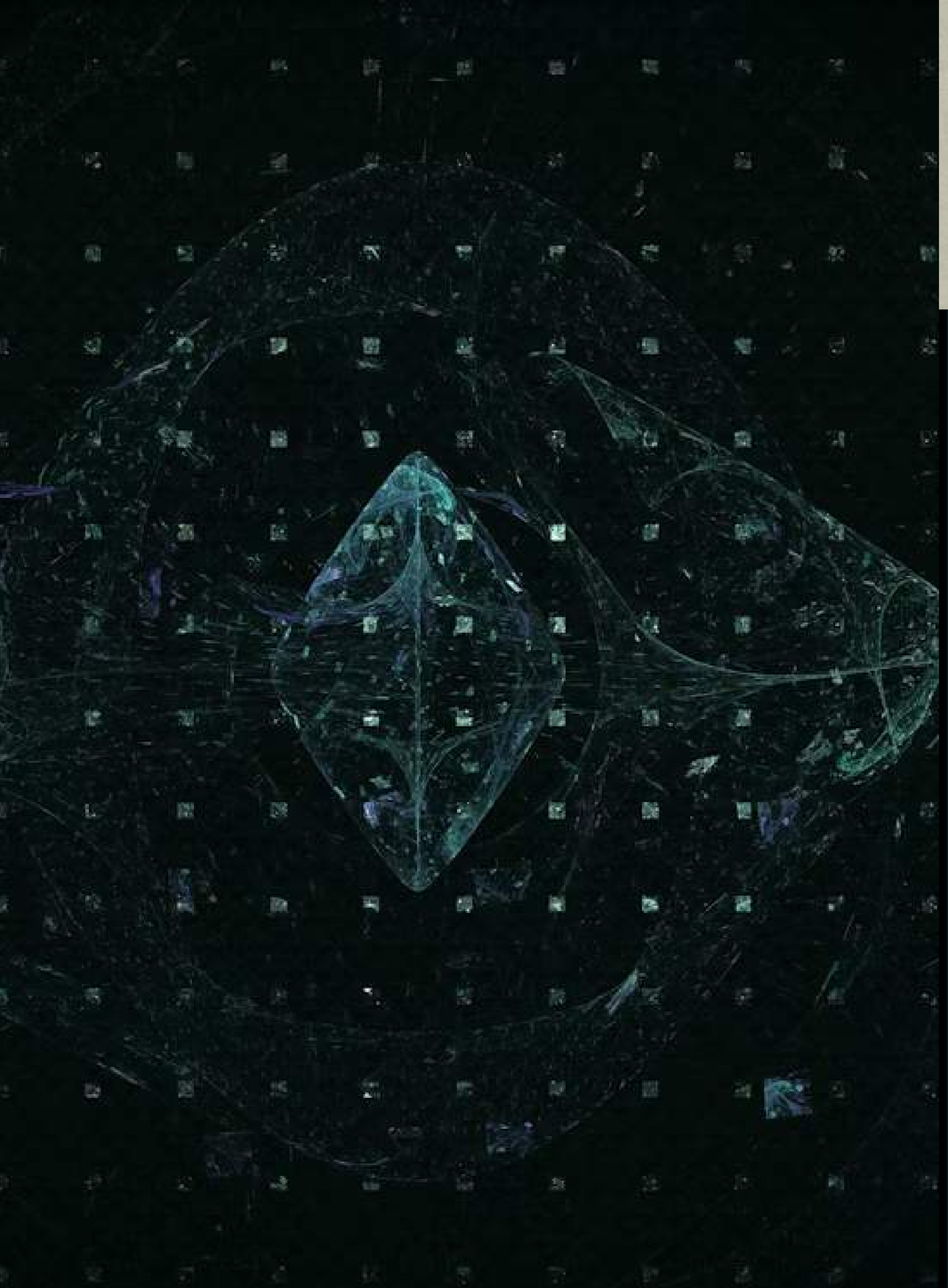

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PROBLEM STATEMENT

- **Financial Impact:** Over $1 billion lost and 46,000+ victims globally due to crypto-fraud.
- **Ethereum Risk:** Accounts for nearly 9% of all global cryptocurrency fraud cases.
- **Anonymity Gap:** Blockchain is transparent but users are anonymous, making it easy for scammers to hide.
- **System Failure:** Traditional rule-based methods are too slow and rigid for evolving attacker techniques.

INTRODUCTION

Ethereum Fraud Detection: A supervised machine learning framework designed to secure the blockchain ecosystem.

Purpose: Automate the identification of fraudulent accounts and suspicious transaction patterns to enhance security.

Target Users:

- Financial Institutions: Real-time monitoring of high-risk transactions.
- Investors: Identifying malicious wallet addresses before interaction.
- Developers: Integrating fraud-detection layers into decentralized apps (dApps).

SOLUTION & KEY FEATURES

FRAUD DETECTION

- Classifies transactions as Fraud or Normal
- Tracks behavior like transaction patterns & balance changes

PREDICTIVE MODEL

- Uses Logistic Regression for fast, accurate decisions
- Detects hidden fraud patterns in data

DATA PROCESSING

- Cleans data & handles missing values
- Converts categorical data into numerical form

SCALABLE SYSTEM

- Handles thousands of transactions efficiently
- Adapts to new and evolving fraud threats

TECH STACK

Language:

Python – The core language for data processing and model development.

Data Handling:

Pandas & NumPy – Used for structured data manipulation and efficient numerical computations.

Visualization:

Matplotlib & Seaborn – Employed for Exploratory Data Analysis (EDA) and generating correlation heatmaps.

Machine Learning:

Scikit-learn – Utilized for dataset splitting, feature scaling, and implementing the Logistic Regression model.

Advanced Encoding:

Category Encoders – Implemented Target Encoding to transform complex categorical blockchain data into numerical values.

WORKFLOW

- **Data Loading:** Imported transaction dataset (50+ features)
- **EDA:** Checked structure, missing values & duplicates
- **Data Cleaning:** Removed unnecessary columns & fixed data issues
- **Feature Selection:** Chose important features using correlation analysis
- **Model Training:** Trained Logistic Regression model
- **Model Testing:** Tested model to ensure accuracy & reliability

RESULTS

- **High Accuracy:** Achieved a 99.14% detection rate with a minimal 0.86% error rate.
- **Precision Security:** Secured performance metrics between 98% and 99% for reliable fraud identification.
- **Optimized Analysis:** Efficiently processes over 50 transaction features to isolate suspicious behavioral patterns.
- **Scalable Framework:** A robust Python-based pipeline designed to adapt to evolving blockchain threats.

CONCLUSION

- Delivers a high detection accuracy of 99.14% with a minimal error rate of 0.86%.
- Successfully identifies and distinguishes fraudulent behavior from legitimate Ethereum transactions.
- Achieves robust performance metrics with Precision and Recall scores between 98% and 99%.
- Provides a scalable, data-driven framework to enhance security and minimize financial risk in the blockchain ecosystem.

THANK YOU